

Abstract

A conventional forecast ensemble does not produce extreme events. An ensemble of N members cannot resolve an event whose probability is below about one in N : on average no member lands at the observed value, and the forecast carries no information about the tail. We report results from AI+RES, a rare-event sampling method that runs 64 forecast trajectories and resamples them five times over the forecast. At each resampling step it scores every trajectory by its forecast outcome and prunes the population toward the extreme. The score comes from a second, more efficient AI weather model, which forecasts each trajectory’s outcome from the resampling step to the target lead time; the population is ranked by that forecast outcome. On the seven test cases at or beyond the 96th percentile of the 1959 to 2022 climatological distribution, AI+RES produced forecast members that reached the observed anomaly in six of the seven. Direct ensemble sampling reached it in none of the seven, and the operational Climate Forecast System version 2 (CFSv2) in none of the seven. One AI+RES forecast takes 7.2 to 8.7 hours and 61 to 64 H100 GPU-hours on a single eight-GPU node, about \$156 at \$2.50 per GPU-hour. Obtaining the same set of extreme trajectories by direct sampling is estimated at 5 to 23 times that cost, and the multiple grows with the rarity of the event. Nine test cases support a qualitative assessment while we currently work on a more quantitative result.

1 The problem

Weather forecasts are run as ensembles. The same model is launched many times from slightly different starting points, and the spread of the outcomes is the forecast. The number of runs sets the smallest probability the ensemble can resolve. An ensemble of N runs resolves probabilities down to about one in N and no further: an event rarer than that is, on average, absent from every run, and if one run does land on it, that single run cannot be told apart from chance. The direct ensembles in this brief have 24 members, so their floor is about 1 in 24, a 4% probability. At a lead of three weeks, the time from launch to the event, the events that matter most are the ones below that floor. The 2021 Pacific Northwest heat dome ran 7.72 K above normal over its region for a week, which for that region and season is about a 1-in-22-year week (Table 1). A 24-member ensemble of GenCast, the AI weather model of Section 2, launched three weeks earlier produced no run that reached it. The operational Climate Forecast System version 2 (CFSv2), launched at the same time, forecast a week near normal. Neither model is wrong about the mean state of the atmosphere. Both are limited by the same two things: too few runs to resolve an event this rare, and an ensemble spread that may itself under-represent extremes, so that more runs alone would not be certain to fix it.

2 The method

The algorithm is AI+RES, introduced by [Lancelin et al. \(2026\)](#), whose schematic of the method is reproduced in Figure 1.

Rare-event sampling concentrates a fixed compute budget on the trajectories that approach the event, instead of spreading it evenly over the forecast distribution. We launch 64 forecast trajectories from the same starting conditions. Every three days we stop the trajectories, score each one, discard the low scorers, and duplicate the high scorers in their place. The population stays at 64 members throughout and is progressively concentrated on the trajectories heading toward the extreme. Weights carried alongside each trajectory convert the concentrated population back into an unbiased probability at the end.

The score is the step that decides the outcome. It is a single number assigned to each trajectory at a resampling step, meant to predict how extreme that trajectory’s target week will turn out to be. The population is ranked by it, the lowest-ranked trajectories are discarded and the highest-ranked duplicated, as in panel (a) of Figure 1. The standard choice of score is persistence, the trajectory’s regional temperature anomaly at the moment it is scored.

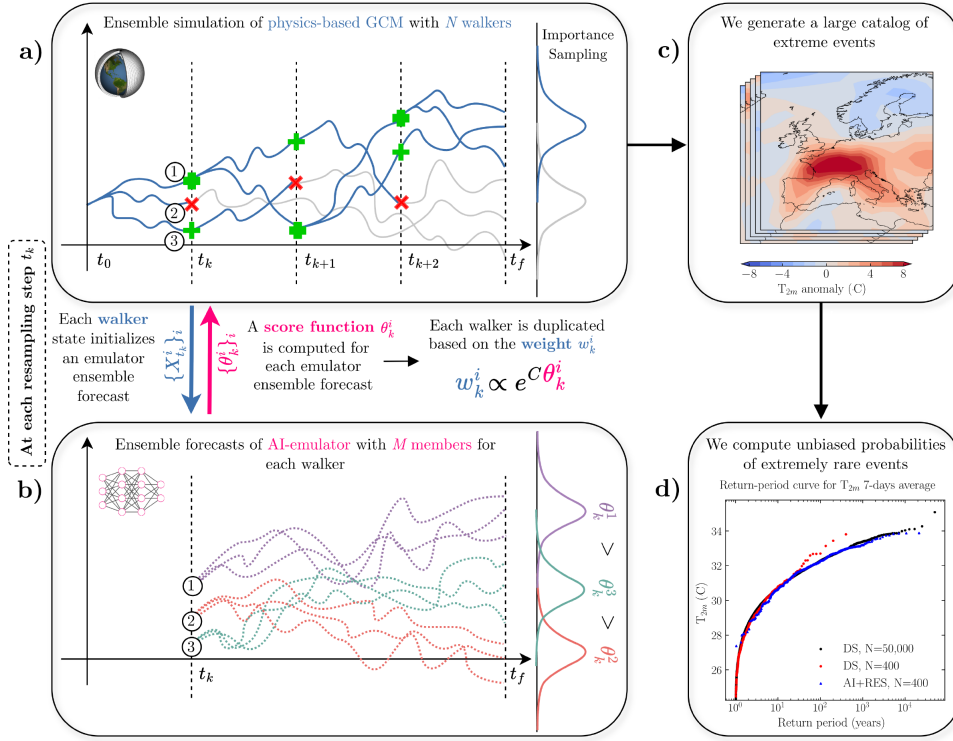


Figure 1: The AI+RES algorithm, reproduced from Figure 1 of [Lancelin et al. \(2026\)](#). **(a)** A population of trajectories runs forward in the physics model. At each resampling time, marked by a dashed vertical, the low scorers are discarded (red crosses) and the high scorers duplicated in their place (green crosses). **(b)** The score comes from an ensemble forecast made by the AI model from each trajectory’s current state out to the target time, which is what carries information the trajectory’s present state does not. **(c)** The run yields a catalogue of extreme events and **(d)** an unbiased return-period curve, at a fraction of the cost of direct sampling (DS in the panel). The configuration reported here follows the same schematic with GenCast as the trajectory model and FourCastNet 3 as the scorer.

Persistence does not tell you what the trajectory does by the end of the forecast. In the 2021 heat-dome run, at the 9-day resampling step, the persistence ranking of the 64 trajectories bears almost no relation to the ranking of their final outcomes: the rank correlation between the two is 0.06, where 1 would be a perfect ordering and 0 none.

We score instead with a forecast, as in panel (b) of Figure 1. A second, more efficient AI weather model is run forward from each trajectory’s current state to the event, and the score is that model’s 6-member ensemble mean of the regional temperature anomaly over the 7 days ending at the event peak. The window is 7 days because that is how long the heat wave itself lasts. Ranked by this score, the same 64 trajectories at the same 9-day step have a rank correlation of 0.80 with their final outcomes, against 0.06 for persistence. That difference decides the result. When the 2021 heat-dome run is repeated with persistence as the score and every other setting unchanged, none of the 64 trajectories reaches the observed anomaly; with the forecast score, 42 of 64 do (Table 3).

The target here is heat waves. A trajectory scores higher the hotter the week it is forecast to produce, so the population is pulled toward heat. The direction of the score is the only part of the method that changes with the kind of extreme. For a cold outbreak the sign is reversed and a trajectory scores higher the colder its forecast week, which is the setting used for Winter Storms Uri and Elliott in Table 1. For a tropical cyclone the score would be built from low mean sea level pressure and high wind speed rather than from temperature.

The trajectory model is GenCast ([Price et al., 2025](#)) at 0.25-degree resolution. The scoring model is FourCastNet 3 ([Bonev et al., 2025](#)). Both were trained on data ending before 2019, so every event tested here is out of sample.

Neither choice is structural. The algorithm needs a trajectory model that can be stopped, checkpointed and restarted, and a scoring model that forecasts to the target lead faster than the trajectory model reaches it. Any weather model meeting those two conditions can take either role, so a more accurate or cheaper model released later can be sub-

stituted for one of ours. The component that would be rewritten is the adapter that translates one model’s state into the other’s input format, not the sampling algorithm.

The demonstration of Lancelin et al. (2026), the one Figure 1 is drawn from, ran an intermediate-complexity climate model as the trajectory model, scored by an AI emulator trained on that same model, and targeted heat waves in the model’s own climate. The work reported here replaces both halves with weather models trained on the observed atmosphere, at 0.25-degree resolution, and targets named events in the observed record.

3 Results

Table 1: Nine test cases, each forecast from three weeks before the event. A_L is the event’s temperature anomaly, the week-long average over the event’s region measured against the 1990 to 2019 normal, in kelvin (1 K = 1.8 °F). The question the table asks is whether each method produced at least one forecast member that reproduced the observed anomaly. Each method column gives that method’s most extreme member, which is the closest any of them came to the event. **Bold** marks a method that reached or passed the observed value. Percentile is the event’s rank against the 1959 to 2022 reanalysis record for the same region and calendar season, with the return period estimated from a generalised extreme value fit in parentheses. Lead times differ by column and are given in Section 3.

		A_L , week-mean temperature anomaly (K)			
event	percentile	observed	AI+RES	CFSv2	direct sampling
<i>Extremes</i>					
S. Texas heat dome, Jun 2023	>99.96 (no analogue)	+5.73	+6.19	+1.19	+4.19
Southwest heat wave, Aug 2020	99.96 (1-in-210 y)	+5.14	+4.31	+1.60	+3.82
California heat wave, Sep 2022	99.71 (1-in-37 y)	+4.71	+6.33	+3.68	+4.70
Pacific NW heat dome, Jun 2021	99.60 (1-in-22 y)	+7.72	+10.29	+3.42	+7.08
CONUS warm week, Dec 2025	98.78 (1-in-10 y)	+3.26	+5.12	+0.22	+3.46
Winter Storm Elliott, Dec 2022	98.41 (1-in-5 y)	−14.42	−17.04	−10.74	−9.83
Winter Storm Uri, Feb 2021	96.80 (1-in-4 y)	−7.40	−10.20	−3.86	−6.51
<i>Mild controls</i>					
CONUS warm week, Aug 2024	94.08 (1-in-4 y)	+1.06	+2.37	+1.35	+1.96
CONUS median week, Nov 2023	60.10 (typical)	+0.21	+4.16	+1.75	+3.14
reached, 7 extremes			6 of 7	0 of 7	0 of 7
reached, all 9			8 of 9	2 of 9	3 of 9

Rarity is what separates the methods. Where the event is rare, AI+RES produces members that reach it and the two alternatives do not. Where the event is ordinary, which is the case for the two controls in Table 1, all three methods reach it. The controls are the check that the difference between methods is a function of rarity and not of one model being more accurate in general.

Lead times are not identical across the columns of Table 1. AI+RES forecasts from a 21-day lead. CFSv2 is also at 21 days, as four operational cycles launched 6 hours apart. Direct sampling is a 24-member GenCast ensemble at a 14-day lead in every row except the S. Texas case, where it is 16 GenCast runs at 21 days. A separate 24-member FourCastNet 3 ensemble at 21 days was run on five of the nine cases and reached the observed value on one of them, the 2021 heat dome, with one member. The direct-sampling column therefore had a one-week shorter lead than AI+RES in most rows, which favours that column, and it still reached none of the seven extremes.

The bound on what this configuration reaches is qualitative. The configuration tested here is 64 trajectories with five resampling steps at a 21-day lead. On the evidence of these nine cases it produces the event for extremes up to about the 99.9th percentile, roughly a 1-in-100-year week for a given region and season. It reached one event with no analogue in the 64-year record, the S. Texas heat dome, and missed another, the Southwest heat wave at an estimated 1-in-210-year return period. That miss is the rarest case in the set by return period. Resampling moved the population 1.75 K toward the observed value and stopped 4.4 K short of it. The boundary for this budget is therefore near the 1-in-100-year level rather than beyond it. Past that boundary the limit is population size, which

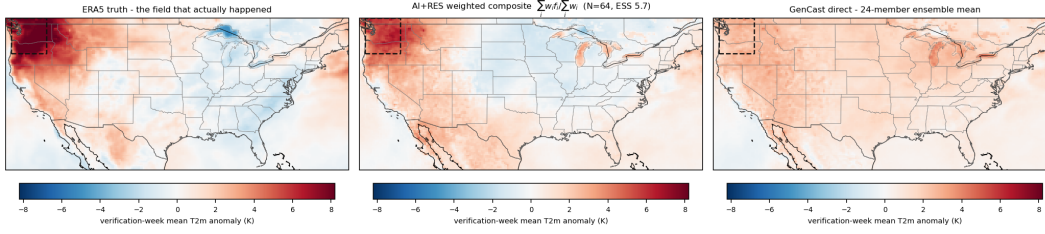


Figure 2: The 2021 Pacific Northwest heat dome, as maps of the week-average temperature anomaly. **Left.** The observed field. **Middle.** The AI+RES population three weeks out, with the dome in the observed location at the observed intensity. **Right.** The 24-member direct GenCast ensemble average from the same starting point, which contains no dome. The dashed box is the scored region. The middle panel is a weighted composite of trajectories selected for this event, so it shows where the method placed its samples and is not a forecast in the ordinary sense.

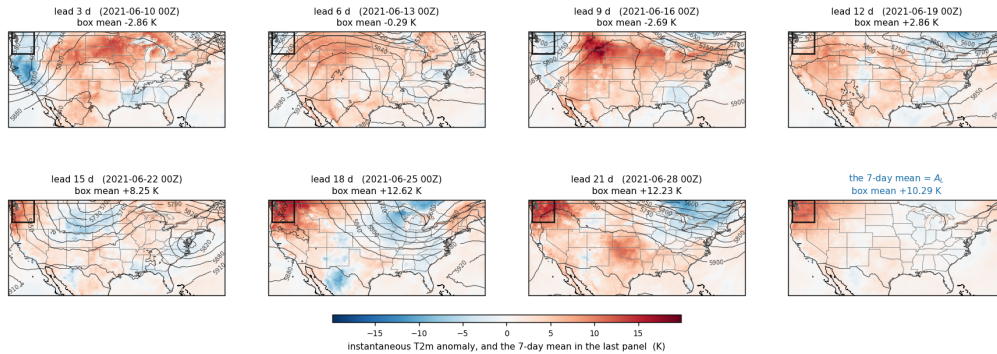


Figure 3: The single hottest surviving trajectory of the 2021 heat-dome run, followed from the 3-day resampling step to the event, with the instantaneous temperature anomaly shaded and the upper-atmosphere pressure pattern contoured. At the 9-day resampling step this trajectory had the fourth-coolest surface temperature of the 64 alive and would have ranked near the bottom on the persistence score. The scoring model ranked it tenth and made eleven copies of it, the most of any trajectory at that step. The contours show why: the upper-level ridge, the high-pressure pattern that produces the heat dome a week later, was already forming over the Pacific just off the coast, while the land surface beneath it had not yet warmed.

scales linearly in compute. With nine cases, that percentile is an estimate from a small sample, not a measured threshold. The seven extremes were chosen as some of the most extreme events of the out-of-sample period, with no formal selection rule.

Figure 2 shows the 2021 heat dome as maps: the AI+RES population reproduces the dome in the observed location and at the observed intensity, and the direct ensemble average from the same starting point contains no dome. Figure 3 follows the hottest surviving trajectory of that run through the resampling steps.

4 Cost and speed

Table 2: One AI+RES forecast, measured, on one node of eight NVIDIA H100 GPUs. Dollar figures use \$2.50 per GPU-hour, or \$20 per hour for the node, a mid-market on-demand cloud rate.

hardware	1 node, 8x NVIDIA H100 (80 GB)
settings	64 trajectories, 5 resampling steps every 3 days, 6 scoring runs per trajectory
forecast horizon	21 days, scored over the final 7-day window
wall-clock, one forecast	7.2 to 8.7 hours
compute, one forecast	61 to 64 H100-hours
cost, one forecast	\$153 to \$160
throughput	about 3 forecasts per node per day

The alternative route to trajectories that reach the event is to run a large direct ensemble until some of its members land there by chance. Table 3 prices that route, from the probability AI+RES assigned each event and the measured

cost of one direct 21-day GenCast run, about 0.4 H100-hours or \$1.00.

Table 3: *Estimated, not measured.* Direct-sampling cost to obtain the same tail sample AI+RES delivered, against the \$156 AI+RES run of Table 2. Each row is Eq. (C1) of Appendix C evaluated at the first two columns. No ensemble of this size was run; the probabilities carry about a factor of two at 64 trajectories, and so do the multiples.

event	members AI+RES put on target	forecast probability, $\hat{p}$	direct runs needed (est.)	direct-sampling cost (est.)	cost multiple
Pacific NW heat dome, Jun 2021	42 of 64	0.054	780	\$780	5x
California heat wave, Sep 2022	25 of 64	0.019	1,320	\$1,320	8x
S. Texas heat dome, Jun 2023	6 of 64	0.0027	2,220	\$2,220	14x
Winter Storm Elliott, Dec 2022	20 of 64	0.0079	2,530	\$2,530	16x
Winter Storm Uri, Feb 2021	32 of 64	0.0090	3,560	\$3,560	23x

The saving in Table 3 is not the primary result. At the \$156 budget of Table 2, direct sampling returns no members at all on these events, which is the measured 0 of 7 in Table 1, most of it at the shorter 14-day lead. The cost multiple grows with rarity, from 5x on the least rare event priced here to 23x on the rarest, because direct-sampling cost scales as the reciprocal of the event probability while AI+RES cost is flat in it. The ordering is not by climatological rarity: the two cold events carry the largest multiples, above the rarer S. Texas heat dome, because the multiple counts the direct runs needed to match the number of members AI+RES put on target, and AI+RES put 20 and 32 members on the cold targets against 6 on S. Texas.

The 2020 Southwest heat wave is absent from Table 3 because neither method reached it. Direct sampling would need at least 3,300 runs, about \$3,300, for a single member.

5 What the result supports

On nine out-of-sample cases, concentrating a fixed compute budget on the trajectories heading toward an extreme produces forecast members on events that direct sampling misses entirely at the same or a larger budget, at the measured cost of Table 2. The result does not support treating these numbers as a calibrated probability service, for three reasons.

1. **Nine events do not establish a hit rate.** The 6-of-7 figure of Table 1 carries a wide interval. A 42-case hindcast campaign designed to produce a reliability curve is built and staged, and has not been run.
2. **The probabilities carry about a factor of two.** They are internally consistent. Where direct sampling can resolve a probability, our estimate agrees with it within sampling error. The absolute level at 64 trajectories is noisy. The method also returns no usable probability for an event milder than the population it has already concentrated, which is a property of the algorithm rather than a failure of a particular run.
3. **The sampled distribution is the forecast model’s, not the atmosphere’s.** The supported claim is that the method reaches the extreme tail of a current forecast model efficiently. Whether that model’s tail is unbiased against the real climate is a separate question, and this work does not settle it.

The 42-case calibration campaign, at an estimated \$1,700 of GPU time for its first 12-case tier, would replace the qualitative bound of Section 3 with a reliability curve. A scoring function for tropical cyclones is needed because the weekly regional average used here averages away a one-day landfall. Lighter model pairs would reduce the per-forecast cost directly, since the current pair requires an 80 GB GPU per trajectory.

References

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A Trajectory plots and anomaly maps for all nine events

These figures show every forecast trajectory of the nine AI+RES runs in Table 1, from the start of the forecast to the event, and beneath each the week-average temperature anomaly as maps. They are not counted in the five pages of the brief. Figure A1 describes the panels, and Figures A2 to A9 use the same layout. The 2021 heat dome comes first because its caption describes the layout, and the remaining events follow the order of Table 1.

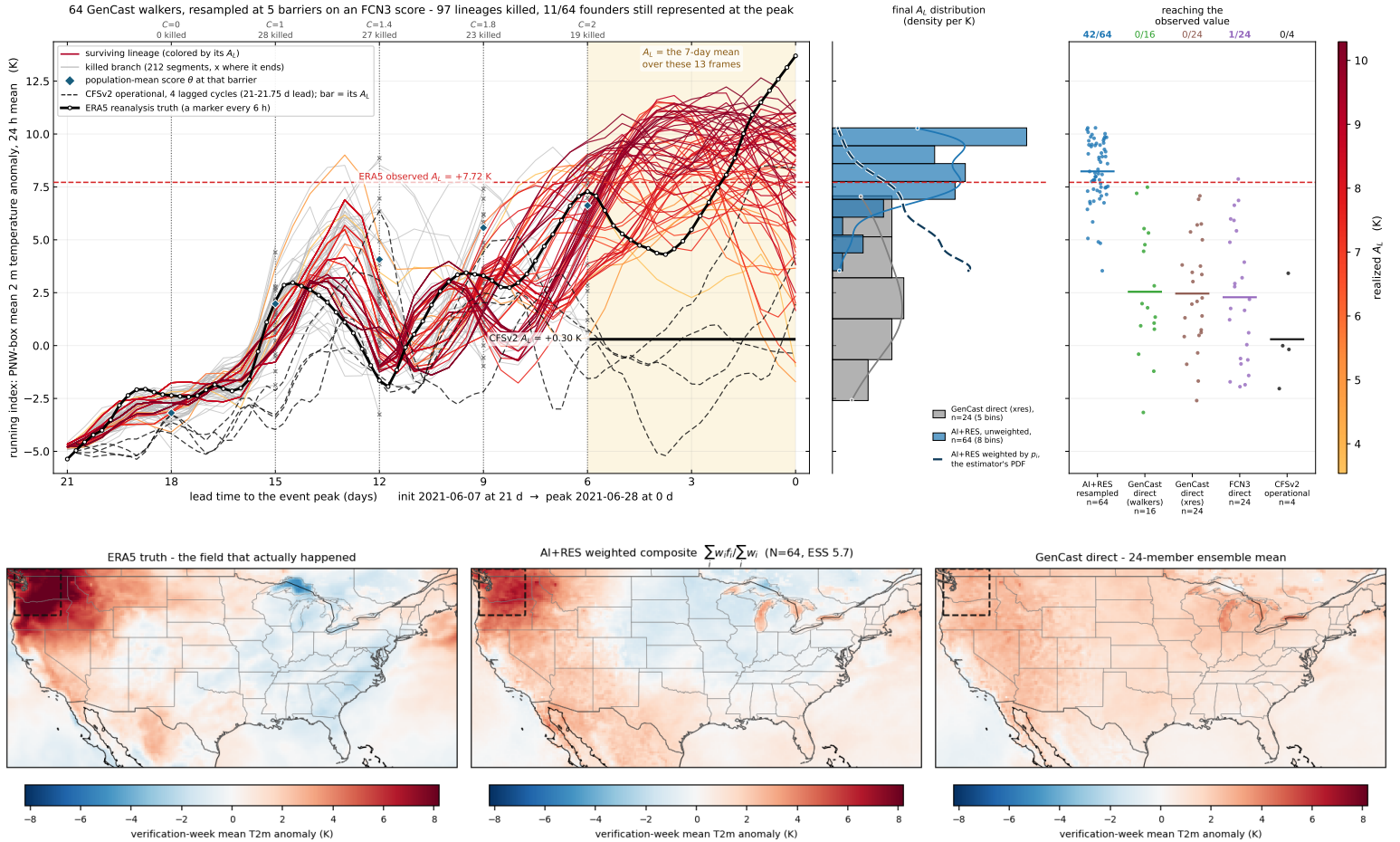


Figure A1: The 2021 Pacific Northwest heat dome, scored over 44 to 50°N and 124 to 118°W, 64 trajectories, forecast from 21 days out. **Top row, left.** Each line is one trajectory, plotted as its running regional temperature anomaly against time. Grey lines were discarded at a resampling step. Coloured lines survived to the event and are shaded by the value they reached. The dotted verticals are the five resampling steps, labelled by how hard the population was pushed and by how many trajectories were discarded. The blue diamonds are the population-average score. The shaded band on the right is the week the forecast is scored over. The black line with markers is the observed weather. The black dashed curves are the four CFSv2 operational cycles at the same lead, with their average drawn as a bar across the scored week. **Top row, middle.** The distribution of final values for the 24-member direct GenCast ensemble in grey, for the AI+RES population in blue, and for the weighted estimate as a dashed line. **Top row, right.** Where each method's members landed, with the number that reached the observed value printed above each column. Where a direct 24-member FourCastNet 3 ensemble was run it appears as a further column. **Bottom row.** The week-average temperature anomaly over the scored week as maps, the observed field on the left, the AI+RES weighted composite in the middle and the direct GenCast ensemble mean on the right, on one shared colour scale. The dashed box is the scored region. The composite shows where the method placed its samples and is not a forecast in the ordinary sense. Observed +7.72 K, reached by 42 of 64 trajectories, most extreme +10.29 K. The direct ensemble reaches it with none of 24 and stops at +7.08 K.

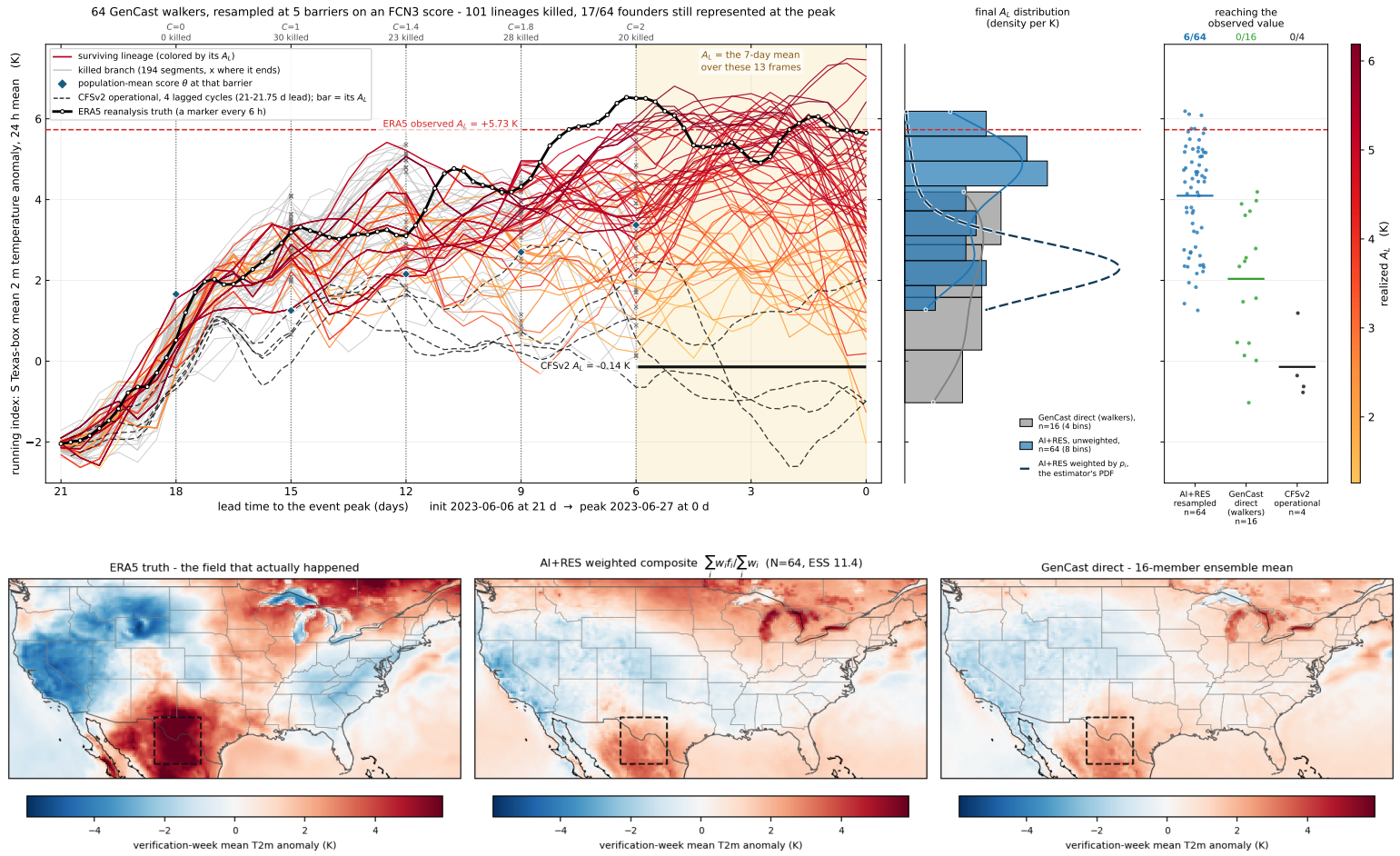


Figure A2: The June 2023 S. Texas heat dome, scored over 26 to 32°N and 106 to 100°W, the event with no analogue in the 1959 to 2022 record. The direct column is the 16 free-running GenCast trajectories at 21 days, the only direct baseline for this region (Section 3). Observed +5.73 K, reached by 6 of 64 trajectories, most extreme +6.19 K. None of the 16 direct trajectories reaches it, the closest stopping at +4.19 K.

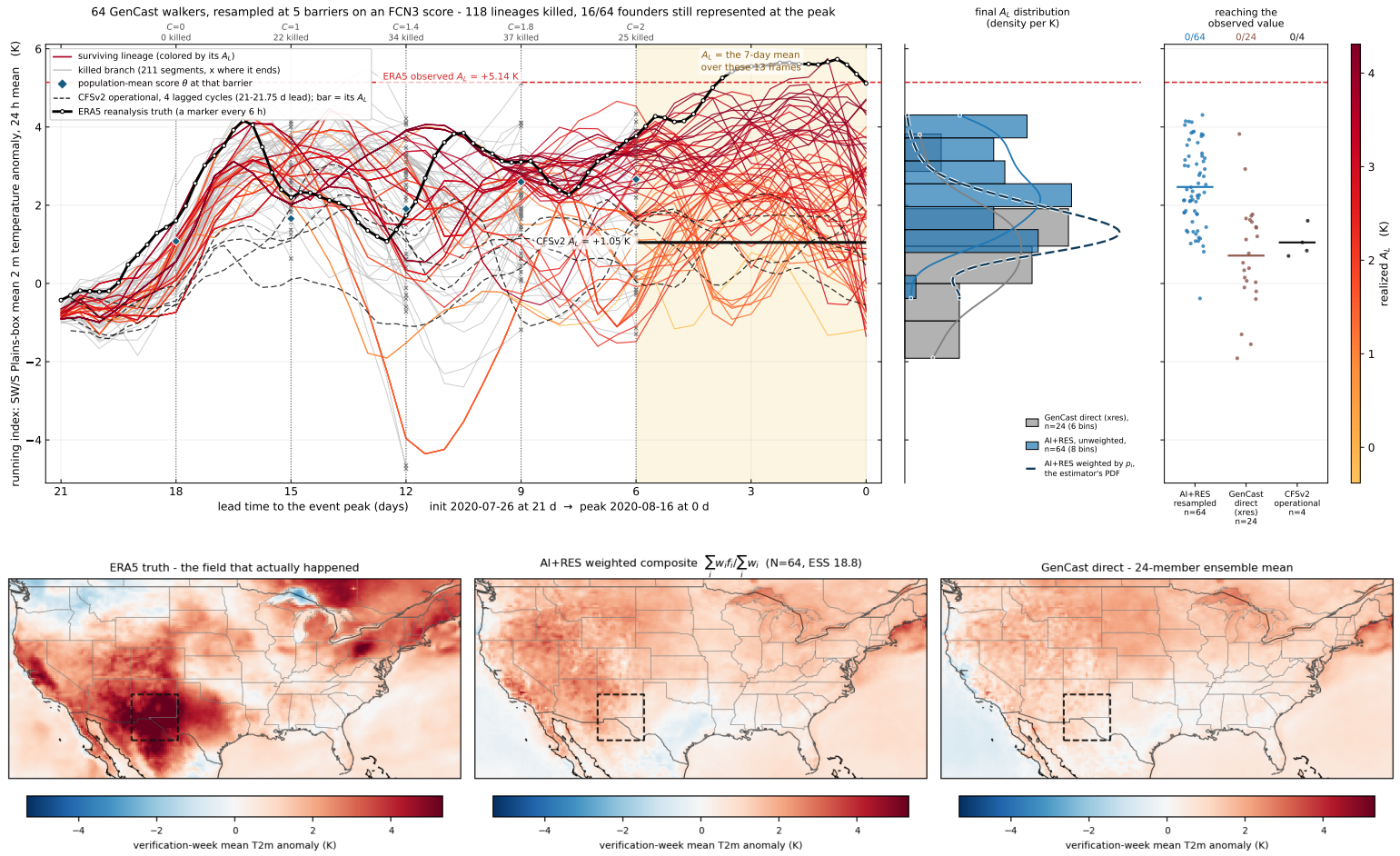


Figure A3: The August 2020 Southwest heat wave, scored over 29 to 35°N and 109 to 103°W, the one target in Table 1 the method does not reach. Resampling moves the population about 1.75 K toward the observed value and ends 4.4 K short of it at +4.31 K. Where direct sampling and the weighted estimate both have members, they agree, so the miss is a limit on how far 64 trajectories can be pushed and not a failure of the weighting.

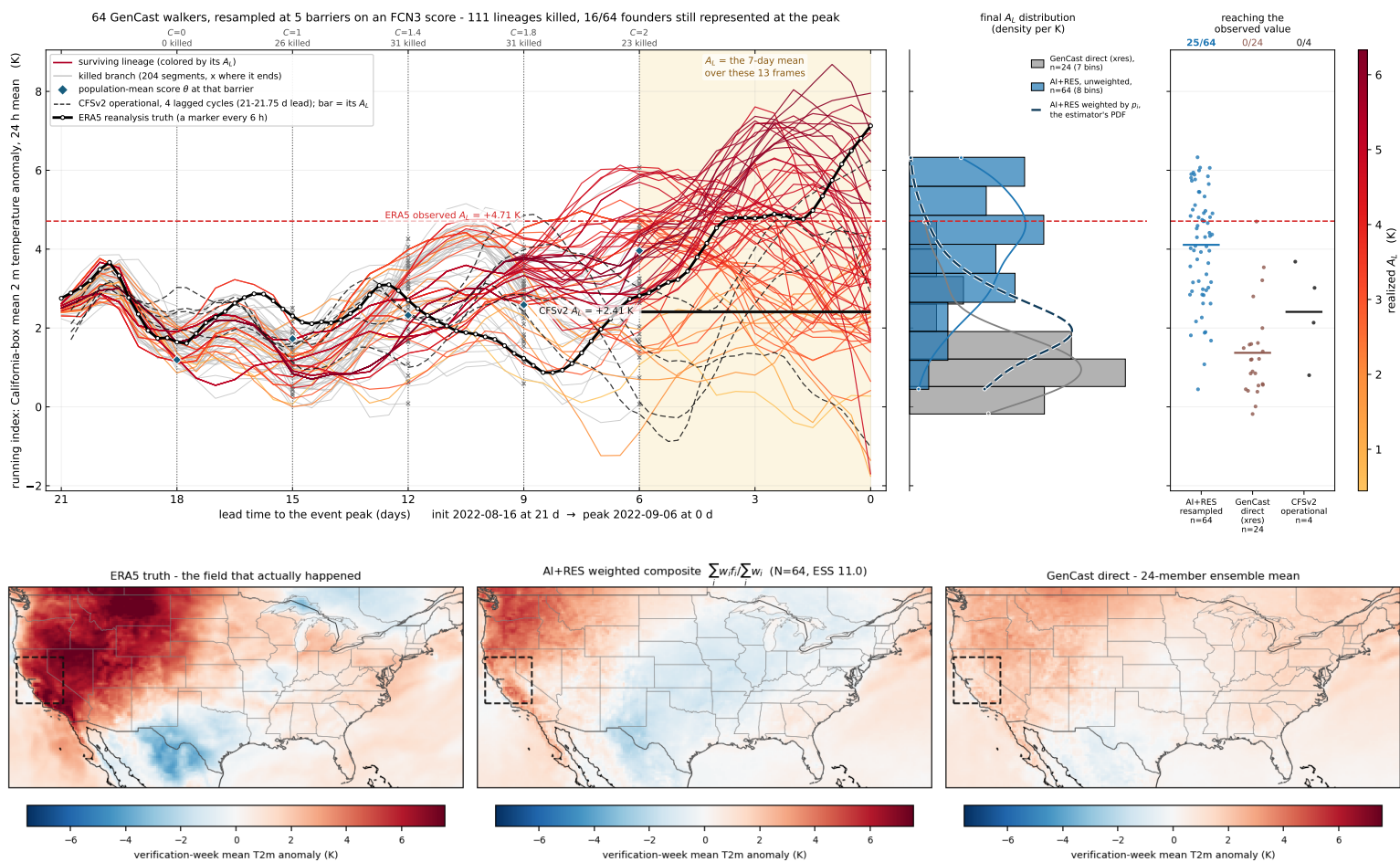


Figure A4: The September 2022 California heat wave, scored over 35 to 41°N and 124 to 118°W. Observed +4.71 K, reached by 25 of 64 trajectories, most extreme +6.33 K. None of 24 direct GenCast members reaches it, the closest stopping at +4.70 K.

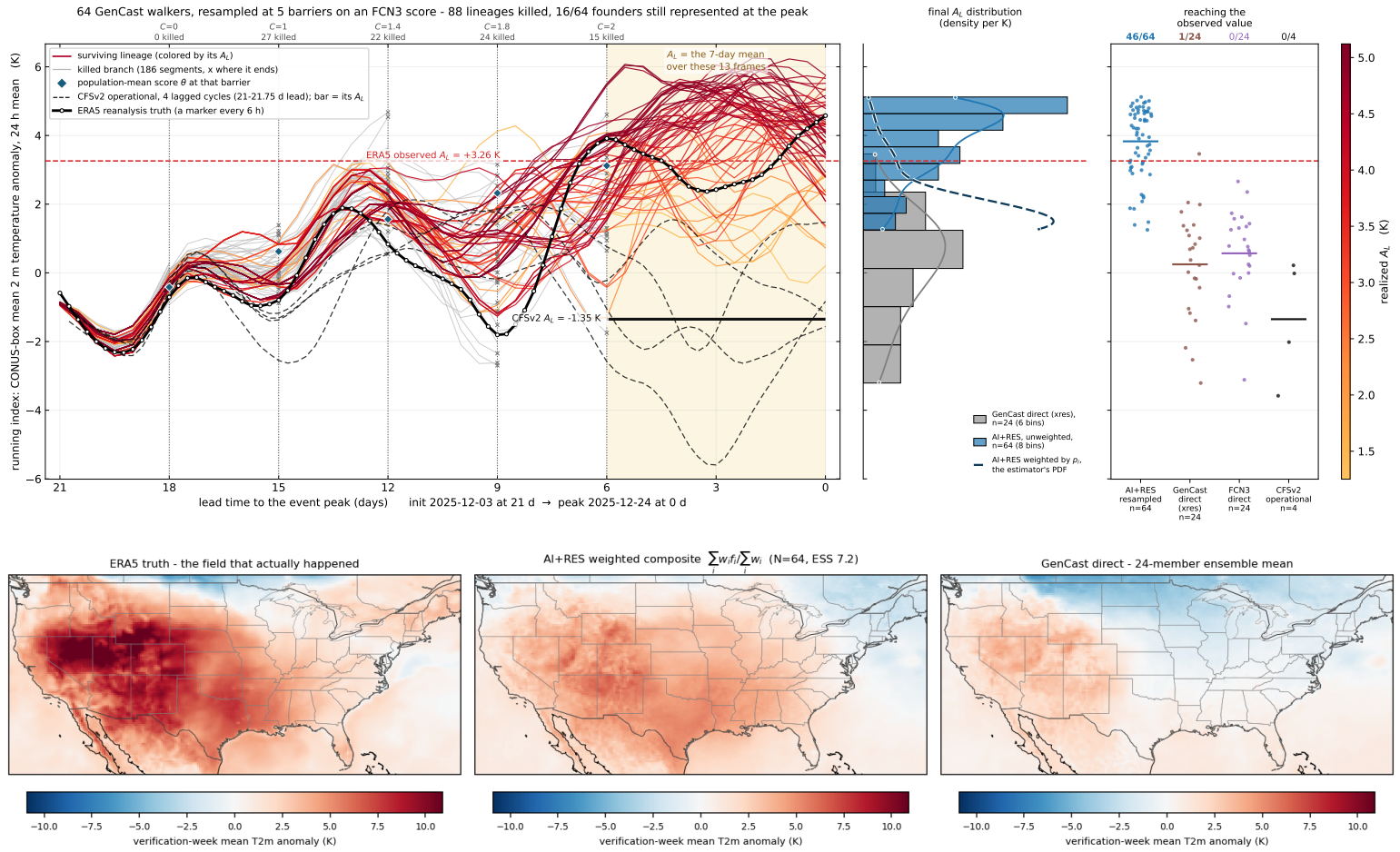


Figure A5: The CONUS-wide warm week of December 2025, scored over 24 to 50°N and 125 to 66°W. This is the one extreme in Table 1 that direct sampling reaches, with 1 of 24 members at +3.46 K. Observed +3.26 K, reached by 46 of 64 trajectories, most extreme +5.12 K. The direct 24-member FourCastNet 3 ensemble at 21 days reaches it with none.

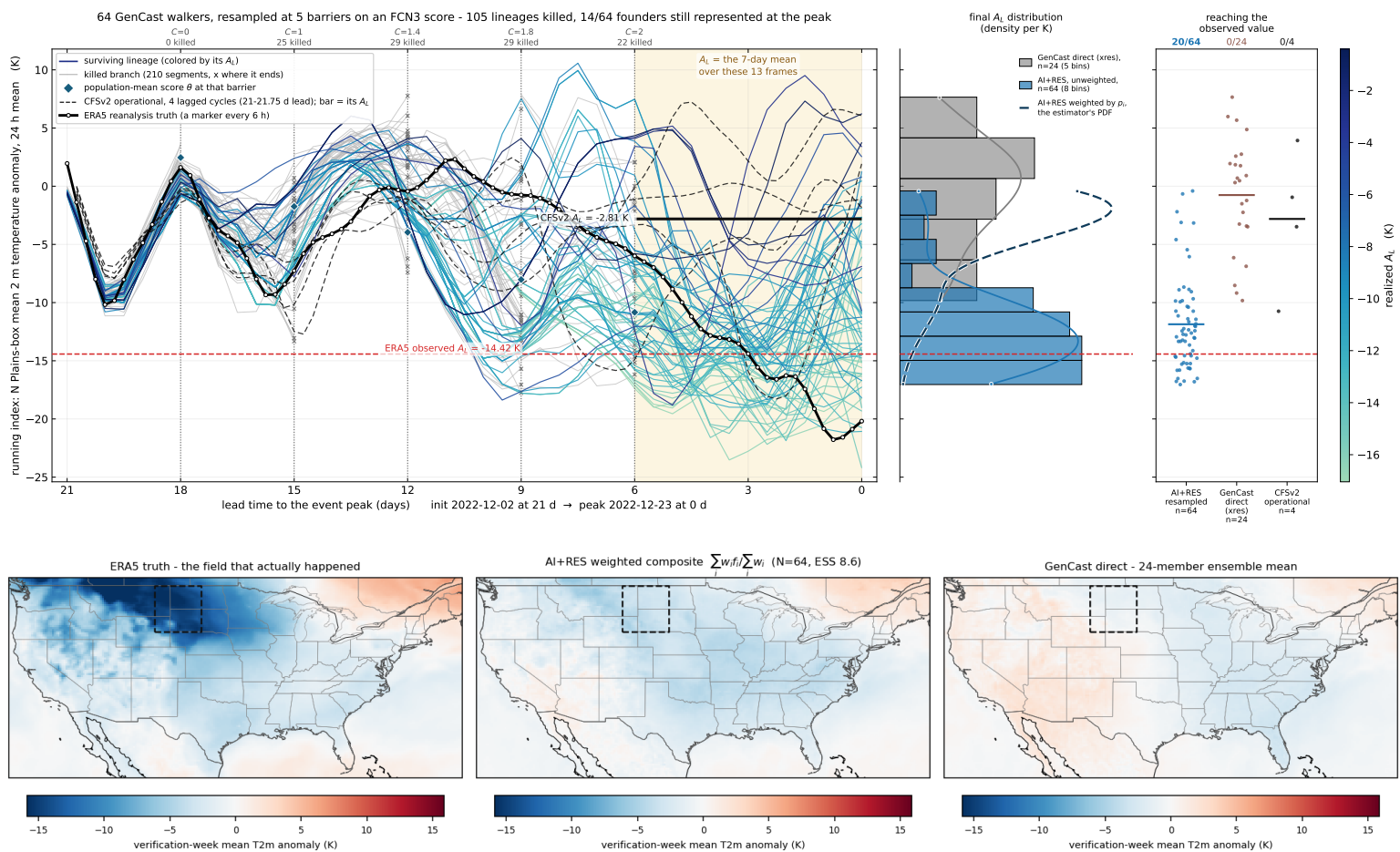


Figure A6: Winter Storm Elliott, December 2022, scored over the northern Plains, 43 to 49°N and 106 to 100°W. This is the deepest cold target in the set relative to the forecast model's own spread. Observed -14.42 K, reached by 20 of 64 trajectories, coldest -17.04 K. The coldest direct member stops 4.6 K short at -9.83 K.

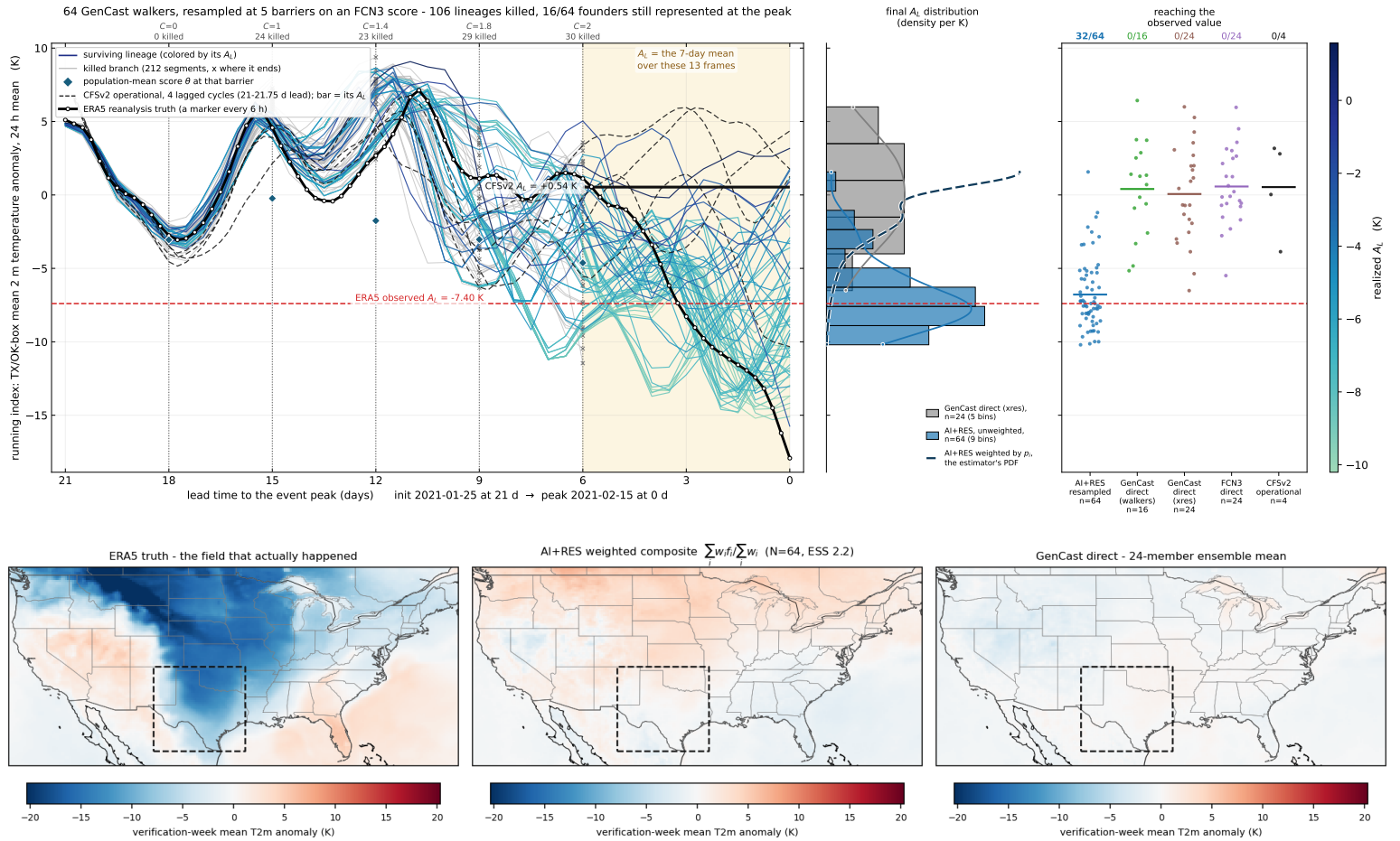


Figure A7: Winter Storm Uri, February 2021, scored over Texas and Oklahoma, 26 to 37°N and 106 to 94°W. The sign of the score is reversed for a cold event, so the method clones the coldest trajectories rather than the warmest. The schedule is otherwise the one used for the heat waves and was not retuned. Observed -7.40 K, reached by 32 of 64 trajectories, coldest -10.20 K. None of 24 direct GenCast members, 16 free-running trajectories or 24 direct FourCastNet 3 members reaches it.

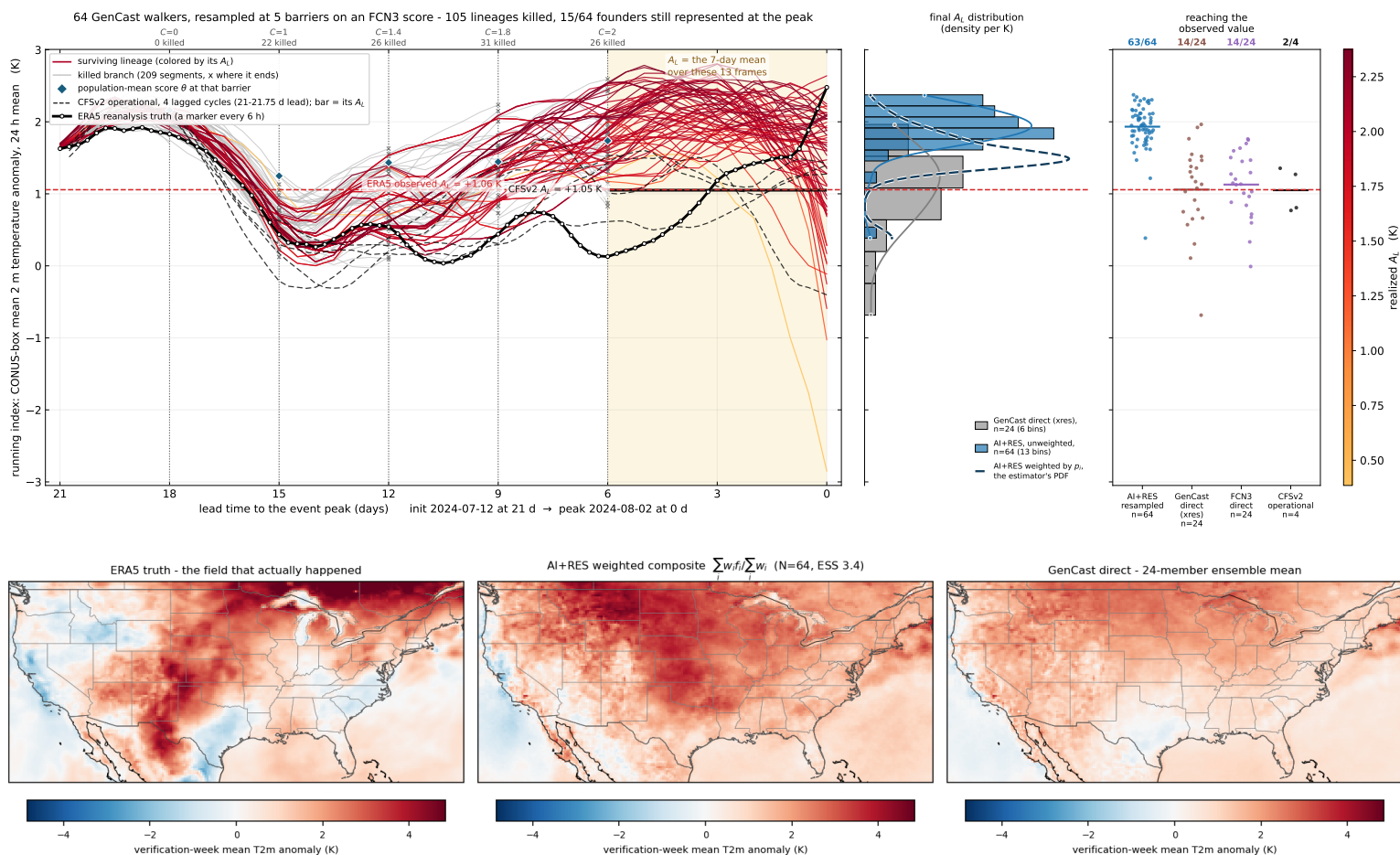


Figure A8: The CONUS-wide warm week of August 2024, the first mild control, scored over the CONUS box of Figure A5. Observed +1.06 K, reached by 63 of 64 trajectories, most extreme +2.37 K. Direct GenCast reaches it with 14 of 24 members, direct FourCastNet 3 with 14 of 24, and CFSv2 with 2 of 4 cycles. Where every method resolves the event, the weighted AI+RES estimate agrees with the direct count within sampling error (Section 5, item 2).

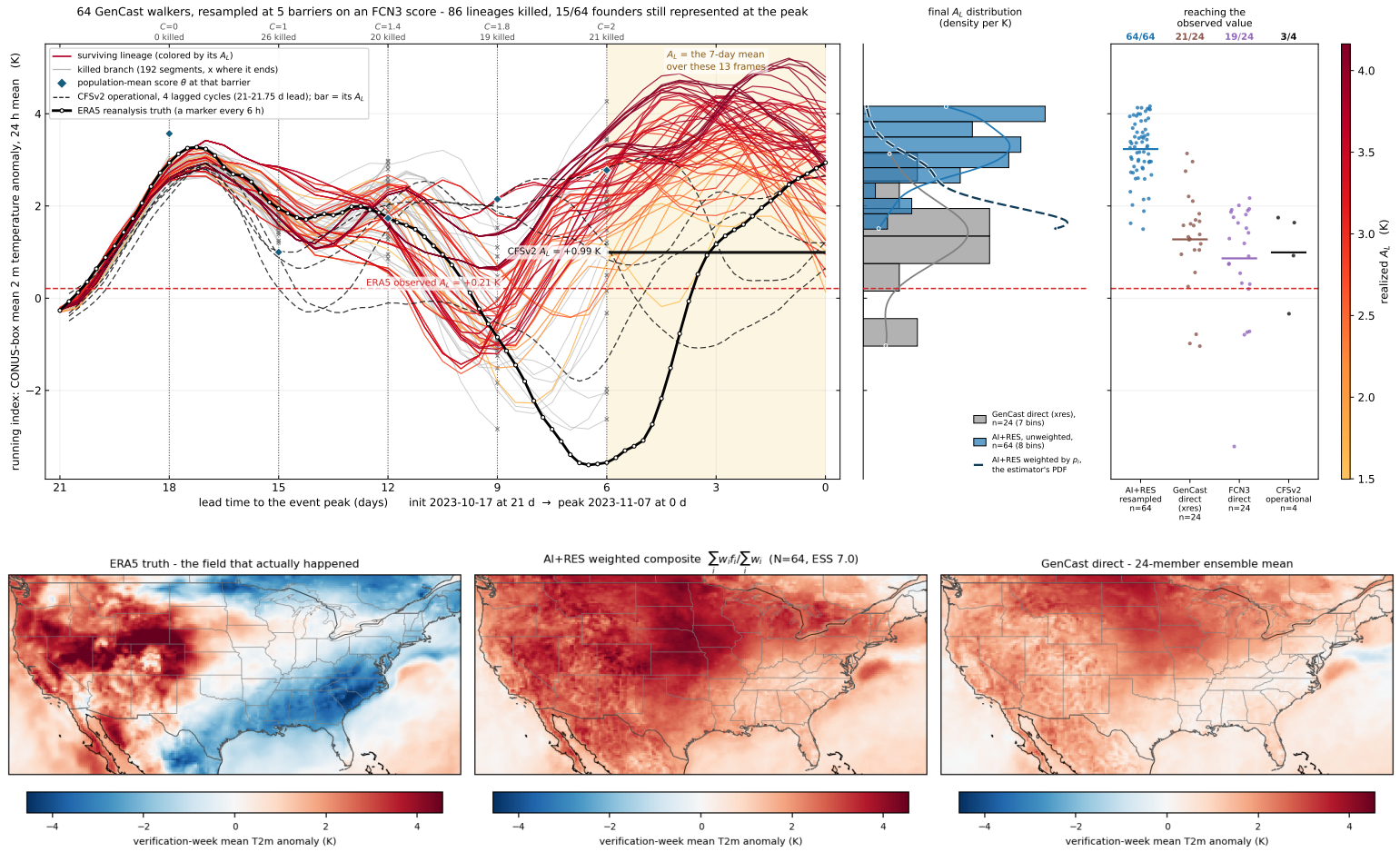


Figure A9: The CONUS-wide median week of November 2023, the second control, scored over the same CONUS box. Observed +0.21 K, reached by all 64 trajectories, most extreme +4.16 K. Direct GenCast reaches it with 21 of 24 members, direct FourCastNet 3 with 19 of 24, and CFSv2 with 3 of 4 cycles. Every trajectory finished past the target, so the population was concentrated entirely beyond it and the weighted estimate carries no information about the event. This is the case referred to in Section 5, item 2.

B Configuration of the AI+RES runs

Table B1 lists every setting of the runs reported in Table 1. The selection strength C sets how hard the population is pushed at each resampling step, through a selection weight $w \propto e^{C\theta}$ on the score θ , the w of Figure 1. These are not the weights of Section 2 that give the final probability. C is 0 at the first resampling step, which leaves the population untouched, and rises as the remaining forecast shortens and the scoring model gains skill over the target week.

Table B1: The AI+RES configuration, unchanged across all nine events of Table 1. The cost of one run at these settings is Table 2.

trajectory model	GenCast 0.25°, trained on data before 2019
trajectory time step	12 h, advanced in segments of 6 steps (3 days)
scoring model	FourCastNet 3, 6-member ensemble per trajectory
trajectories, N	64, held constant through the run
forecast horizon	21 days, initialization to event peak
resampling times	3, 6, 9, 12, 15 days
selection strength, C	0, 1.0, 1.4, 1.8, 2.0, one per resampling step
resampling scheme	diffusion Monte Carlo, pivotal sampling, one fixed seed
score, θ	6-member mean of the area-weighted regional mean of the 7-day-mean 2 m temperature anomaly, sign set by the event
scored window	the 7 days ending at the event peak
scored region	one fixed latitude-longitude box per event, an input to the score, given in the captions of Appendix A
anomaly baseline	ERA5 climatology, 1990 to 2019
trajectory state kept	global, 2 frames, float32

C Derivation of the cost multiple

Write k for the number of AI+RES trajectories that reached the observed value, $\hat{p}$ for the probability AI+RES assigned to the event, the weighted estimate of Section 2, c_d for the cost of one direct 21-day GenCast run, and c_a for the cost of one AI+RES forecast. Each member of a direct ensemble lands at or beyond the observed value with probability $\hat{p}$, so n runs put $n\hat{p}$ members there on average, and $n = k/\hat{p}$ runs are needed for that average to equal k . The cost of those runs is $n c_d$, and the cost relative to one AI+RES forecast is

$$\frac{n c_d}{c_a} = \frac{k c_d}{\hat{p} c_a}, \quad c_d = \$1.00, \quad c_a = \$156. \quad (\text{C1})$$

Both costs are measured. c_d is 0.4 H100-hours per 21-day run at \$2.50 per GPU-hour, and c_a is the range in Table 2 rounded to its midpoint. The 2021 Pacific Northwest heat dome gives the smallest multiple in the set. There $k = 42$ and $\hat{p} = 0.054$, so $n = 780$ runs at \$780, which is $780/156 = 5.0$ times the AI+RES cost. Winter Storm Uri gives the largest. There $k = 32$ and $\hat{p} = 0.0090$, so $n = 3,560$ runs at \$3,560, which is $3,560/156 = 22.8$ times. The five events priced in Table 3 span that 5x to 23x range.

Only $\hat{p}$ is estimated rather than measured, which is why Table 3 is marked as an estimate. At 64 trajectories $\hat{p}$ carries about a factor of two, and the multiples carry the same factor. No ensemble of the sizes in the fourth column of Table 3 was run.